

A New Method for Characterizing CMP's Localized Removal Laws Using a Die-Scale Modeling Approach with Measured Surfaces

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Outline

1. Introduction

- Background, motivation, and research objective

2. Computational modeling

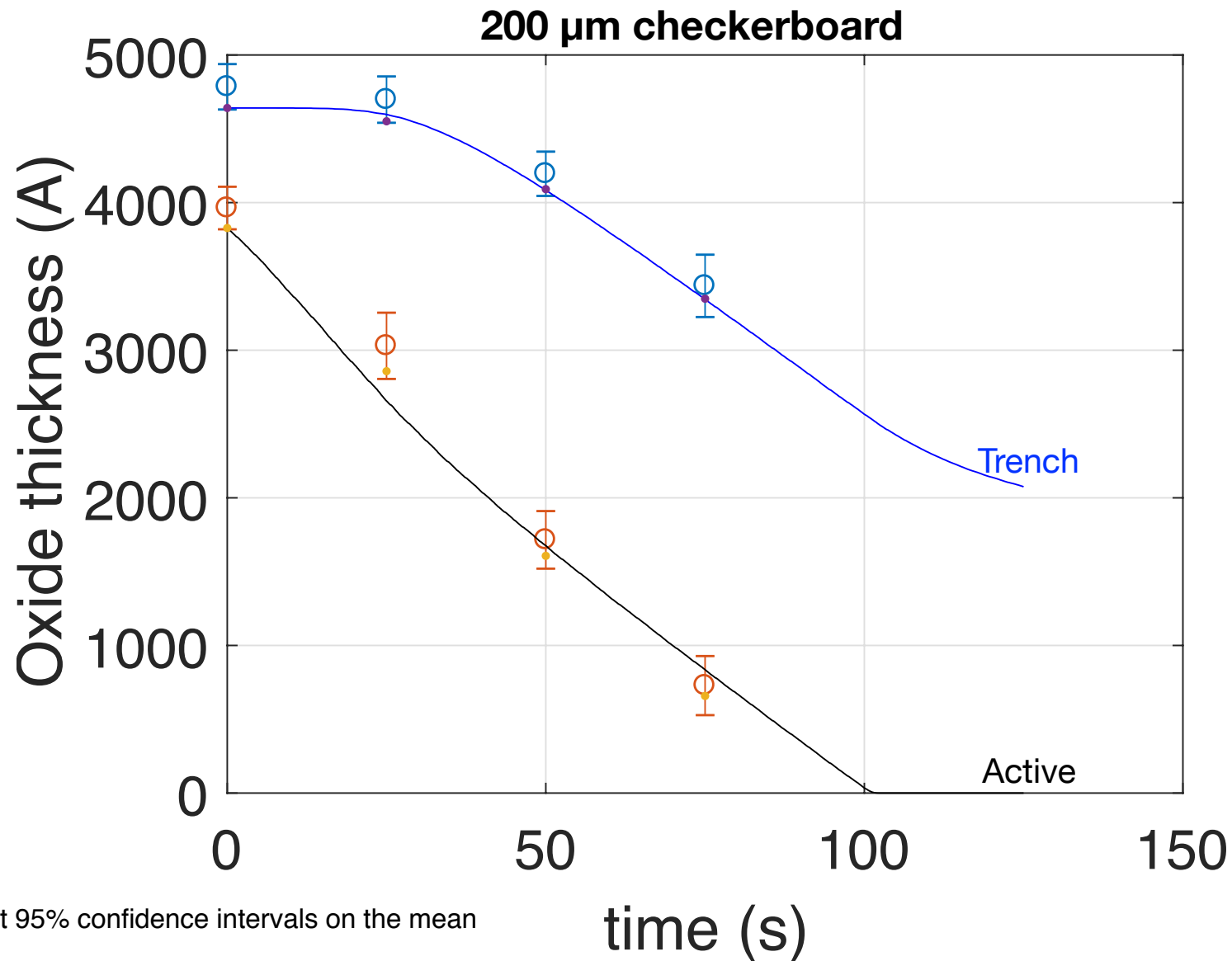
- Calculating the pressure between the pad and die
- Incorporating the role of pad surface asperities

3. Results

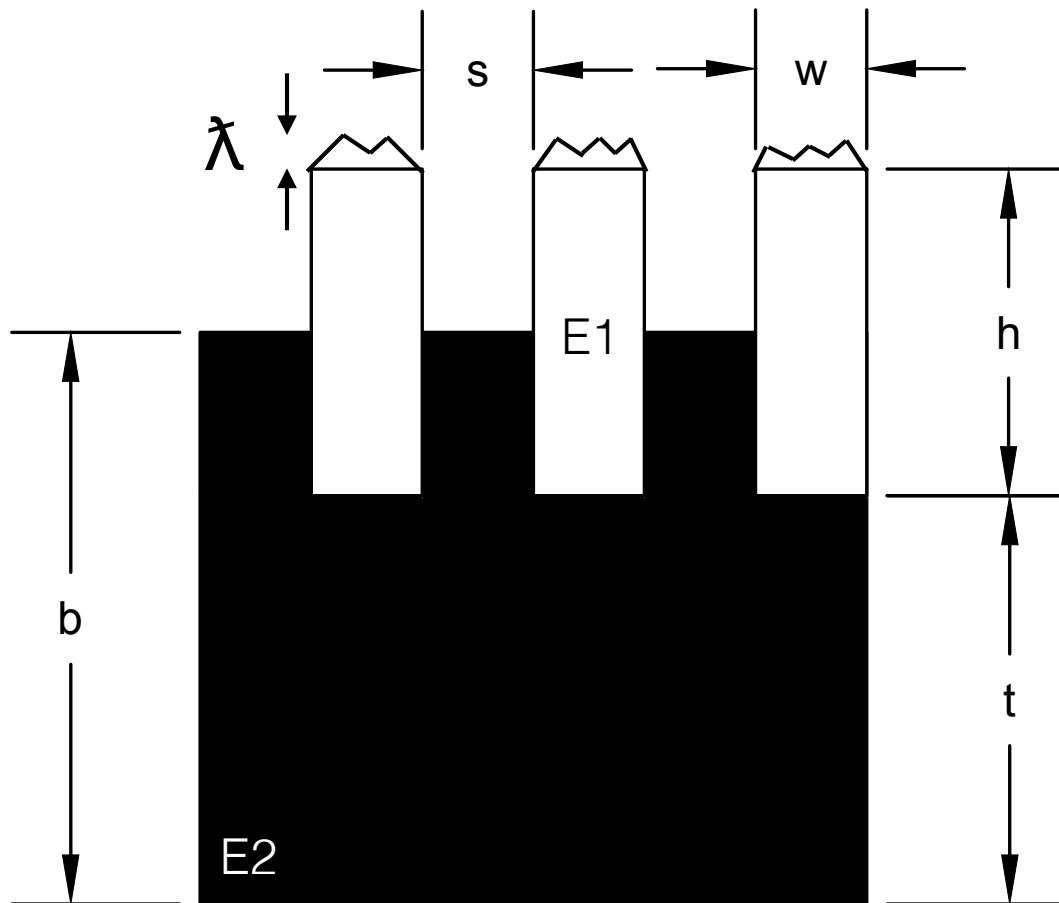
- Feature size, surface texture, and pad segment size dependence

4. Conclusion

Goal: predict oxide thickness vs time for a given pad design



We seek to predict the polish behavior of segmented CMP pads. I parametrize the pad design space with 6 geometric and 3 mechanical parameters



Parameter	Range
E1	500 - 5000MPa
E2	1 - 100MPa
w	0.25 - 3 mm
s	0.25 - 3 mm
h	0.25 - 1 mm
t	0.25 - 2 mm
b	0.5 - 2.5 mm
λ	10 - 100 μm

Identifying the process' removal rate law is essential for accurate simulations but process characterization is complex

Preston's equation

$$\bar{R} = k \cdot \bar{v} \cdot \bar{P}$$

Localized Preston's equation

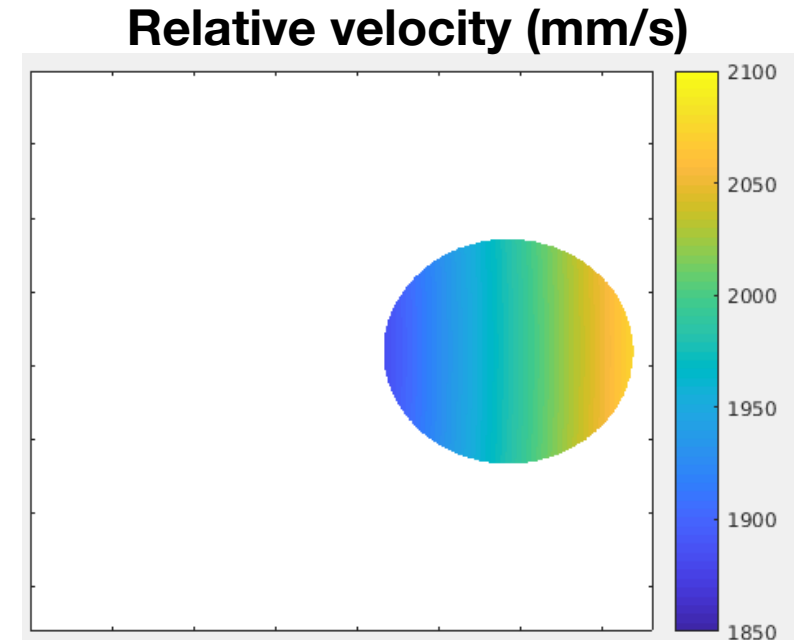
$$R(x, y) = k \cdot v(x, y) \cdot P(x, y)$$

Localized and Generalized Preston's equation

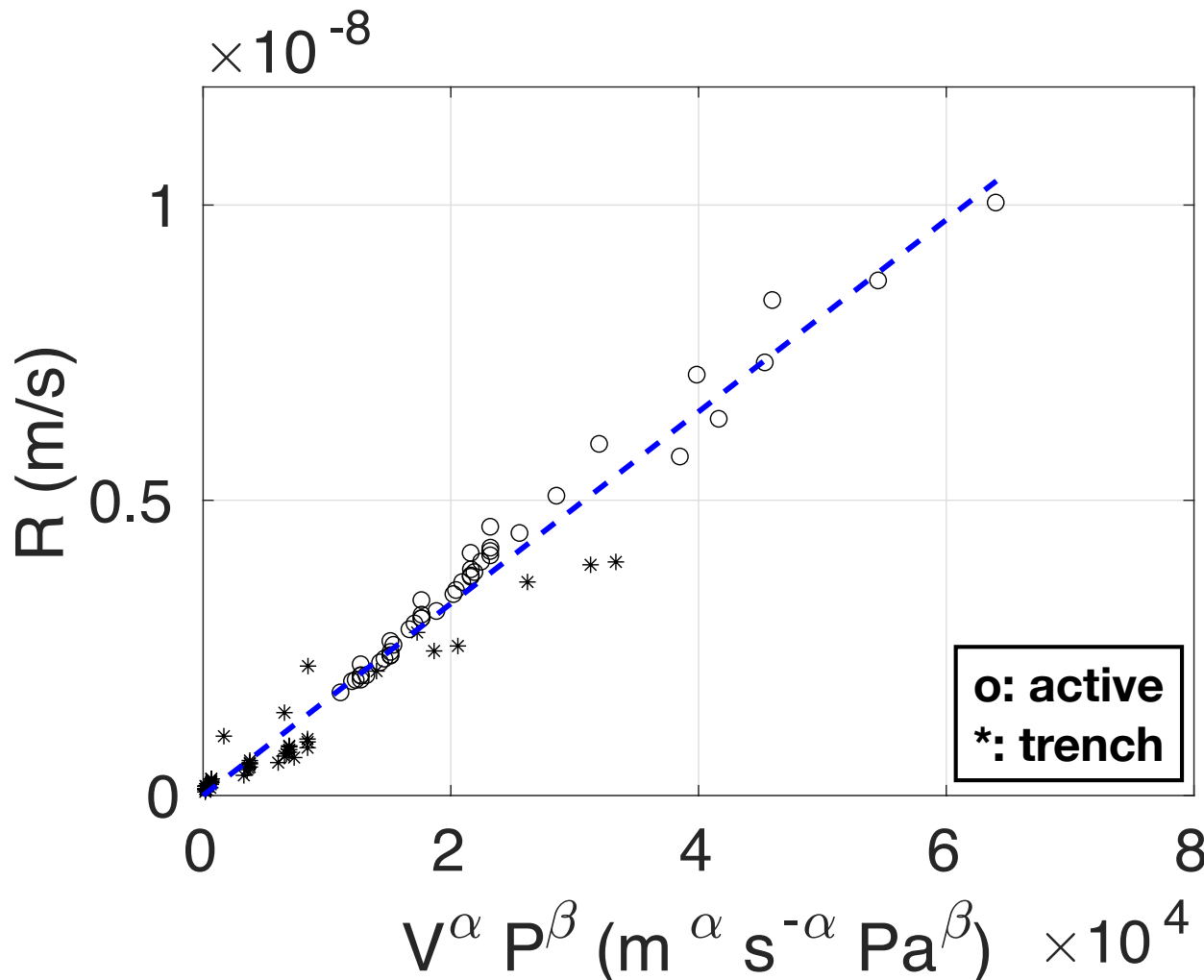
$$R(x, y) = k \cdot v(x, y)^\alpha \cdot P(x, y)^\beta$$



The pressure distribution must be simulated using rough pad surfaces



By using the simulated pressure distributions in conjunction with experimentally measured removal rates (from patterned dies), we can characterize the removal



$$R(x, y) = k \cdot v(x, y)^\alpha \cdot P(x, y)^\beta$$

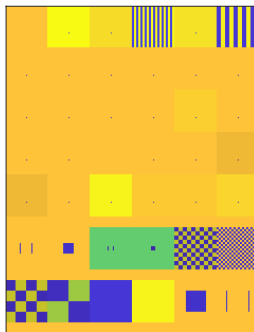
$$k = 1.6 \times 10^{-13} \left(\frac{\text{m}^{1-\alpha}}{\text{s}^{1-\alpha}} \text{Pa}^{-\beta} \right)$$

$$\alpha = 0.39$$

$$\beta = 1.04$$

Method: Use the contact mechanics model to calculate the average pressure across the die, averaged across multiple pad/wafer configurations

Die
topography



Iteration #

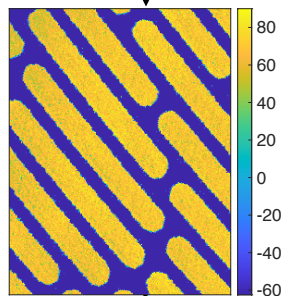
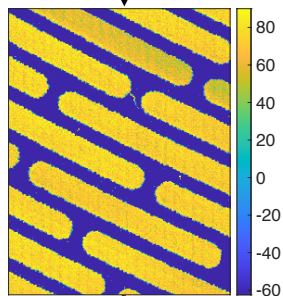
1

2

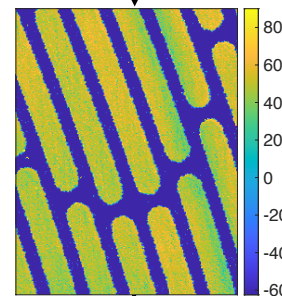
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N

Pad
topography
(μm)

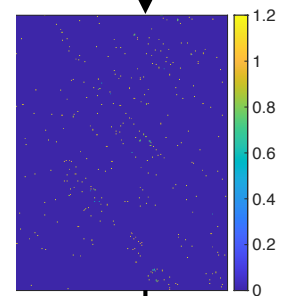
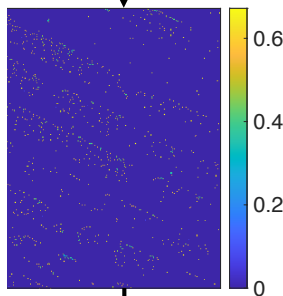


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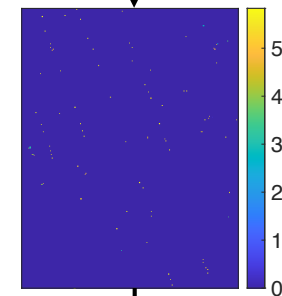


Contact
simulation

$P_i(x, y)$
(MPa)



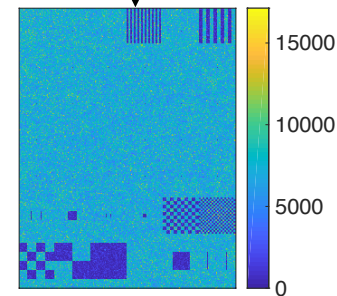
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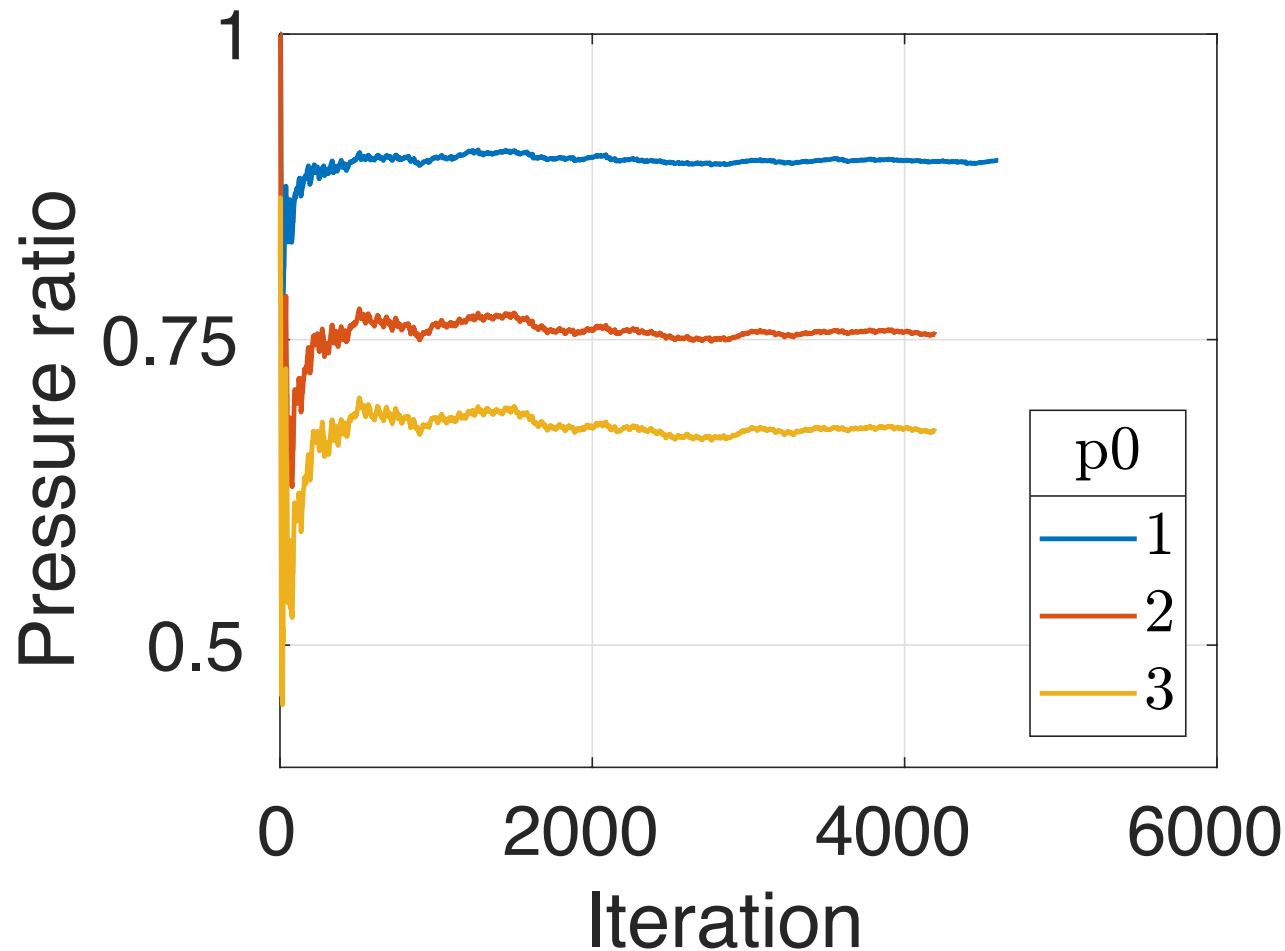
Average

$P(x, y)$
(Pa)

Computational modeling

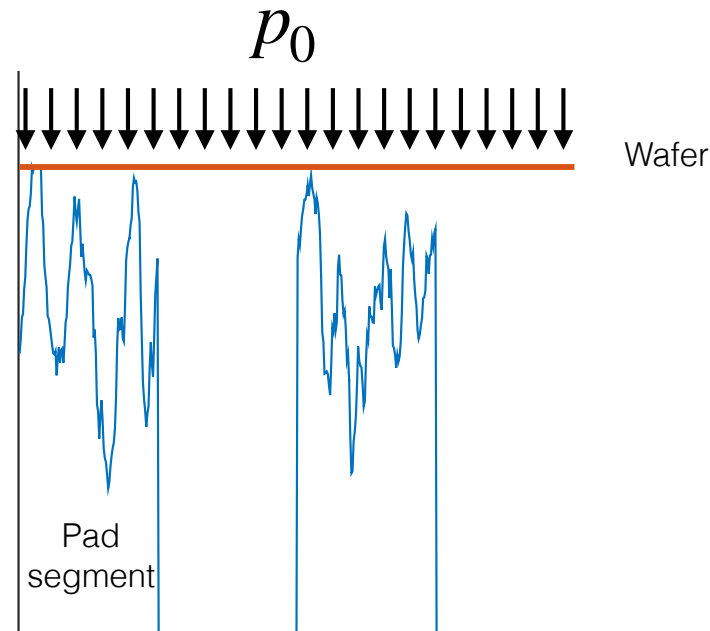


We have found that simulating 2000 different pad/wafer relative configurations is sufficient for convergence

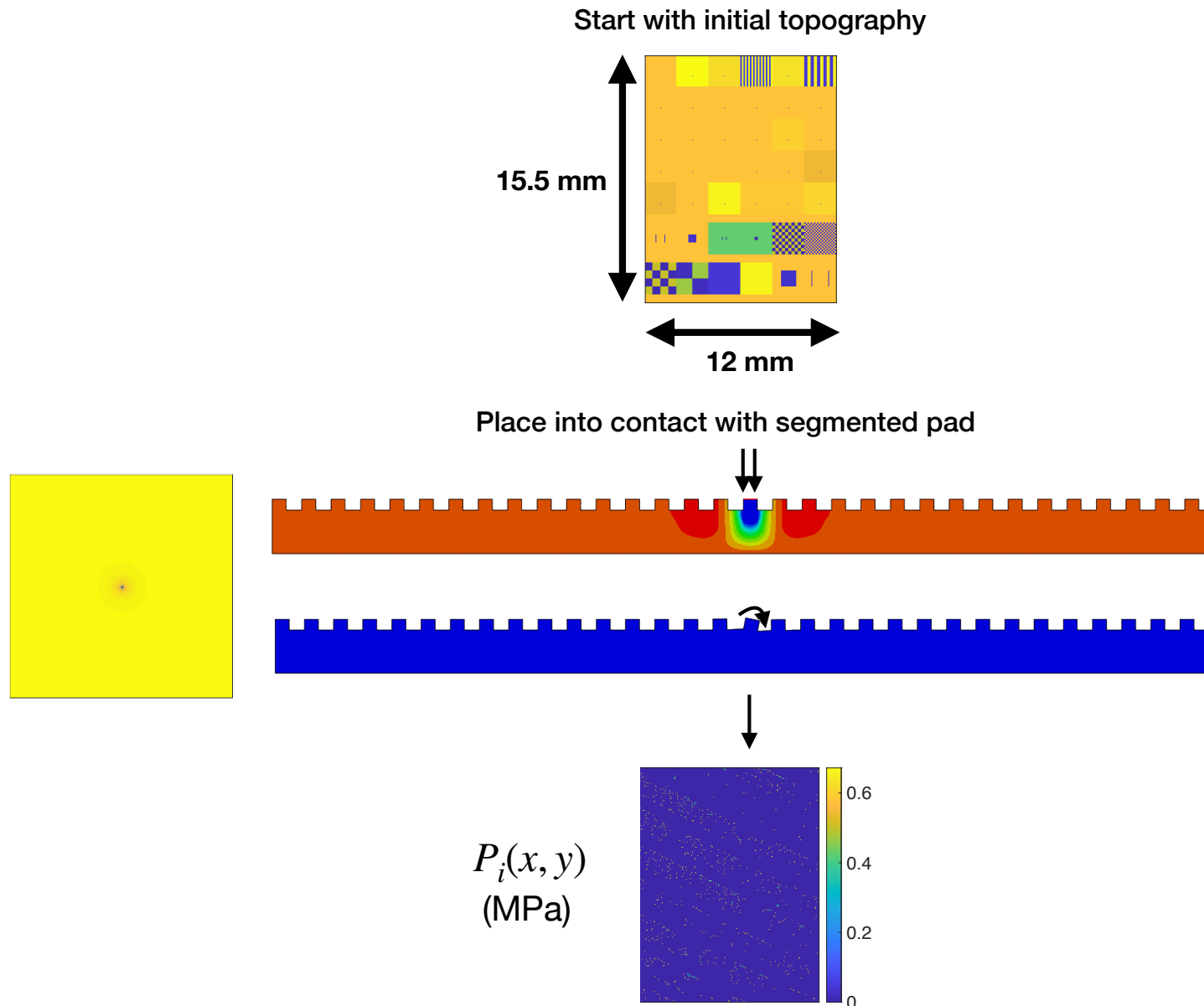


Pressure ratio: $P.R. = 1 - \frac{P_{\text{trench}}}{P_{\text{active}}} \neq P.E. = 1 - \frac{R_{\text{trench}}}{R_{\text{active}}}$

A solid-solid contact model: I am analyzing the mechanical stresses at the interface between pad asperities and the wafer

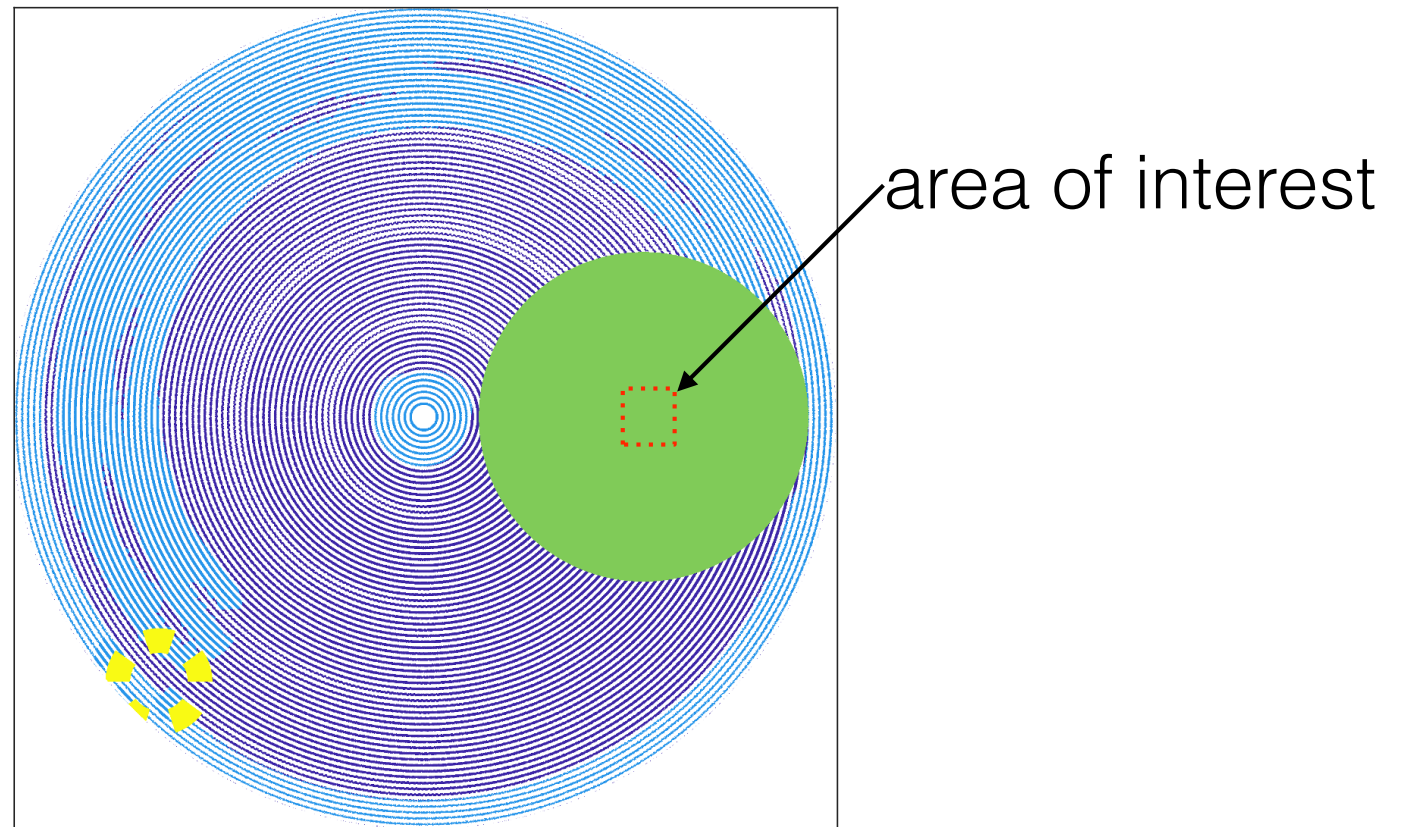


We model the CMP process using a contact mechanics approach that accounts for both bulk and surface pad deformation



Computational modeling

We'll focus on the die that is at the very center of the wafer

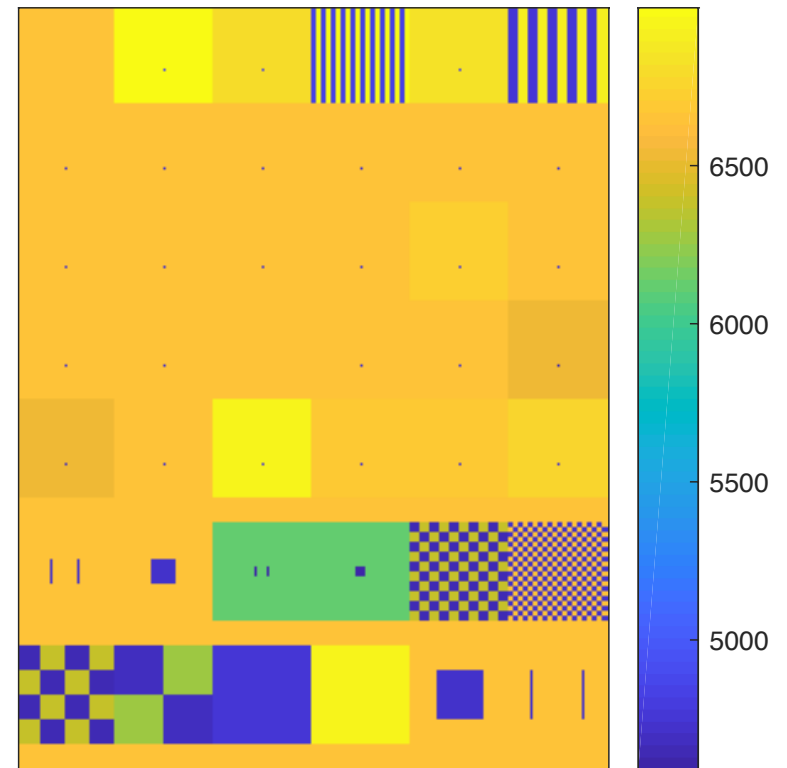


Dies contain arrangements of multiple types of features

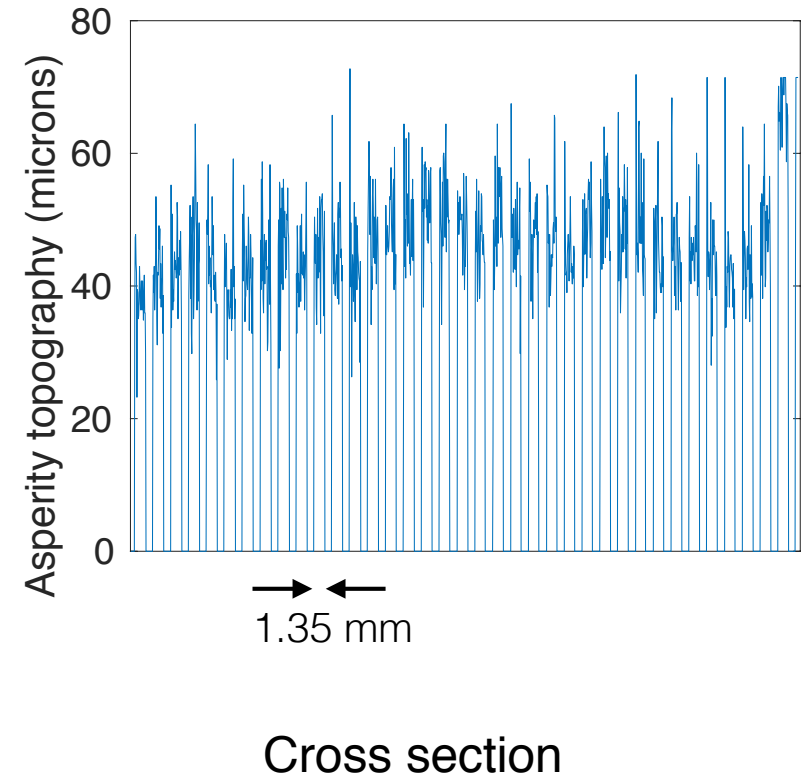
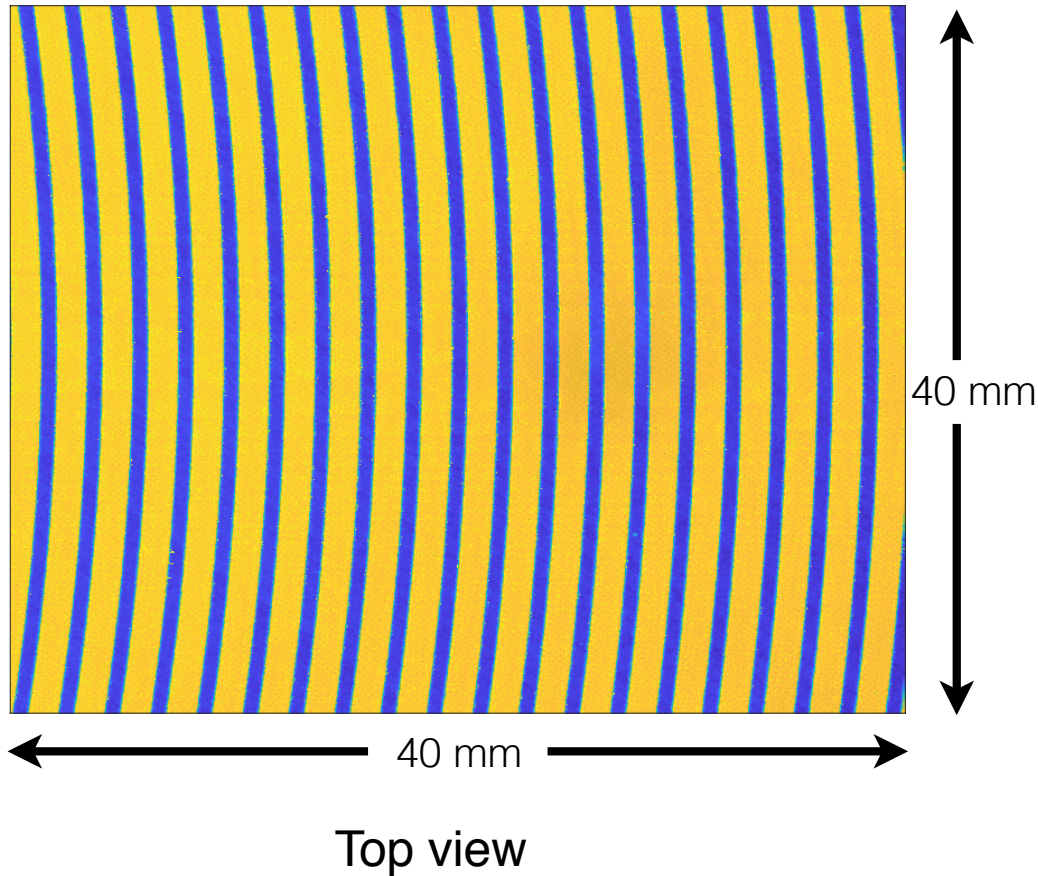
Standard test pattern

	A	B	C	D	E	F	
7	50x50 μm checkerboard	50% 10 μm pitch L/S	50% 0.4 μm pitch L/S	50% 200 μm pitch L/S	50% 4 μm pitch L/S	50% 400 μm pitch L/S	7
6	50% 20 μm pitch L/S	50% 1.0 μm pitch L/S	50% 2 μm pitch L/S	50% 0.2 μm pitch L/S	40% staggered blocks (DRAM)	90% 1.0 μm pitch L/S	6
5	40% staggered blocks (DRAM)	30% staggered blocks (DRAM)	70% 1.0 μm pitch L/S	30% 1.0 μm pitch L/S	50% 1.0 μm pitch L/S	50% staggered blocks (DRAM)	5
4	70% staggered blocks (DRAM)	60% staggered blocks (DRAM)	50% 100 μm pitch L/S	10% 1.0 μm pitch L/S	50% 0.2 μm pitch L/S	20% staggered blocks (DRAM)	4
3	80% staggered blocks (DRAM)	50% 0.9 μm Line / vary Space	10% 0.9 μm Line / vary Space	90% 0.9 μm Line / vary Space	70% 0.9 μm Line / vary Space	30% 0.9 μm Line / vary Space	3
	40% staggered blocks (DRAM)						
2	500x500 μm active block 40% DRAM	500x500 μm trench block 40% DRAM	200x200 μm active block 40% DRAM	200x200 μm trench block 40% DRAM	200x200 μm checkerboard	100x100 μm checkerboard	2
	40% staggered blocks (DRAM)						
1	500x500 μm checkerboard	1000x1000 μm checkerboard	2000x2000 μm trench only	2000x2000 μm active only	1000x1000 μm trench block 40% DRAM	1000x1000 μm active block 40% DRAM	1
	40% staggered blocks (DRAM)						

Topography ($\text{\AA}$)



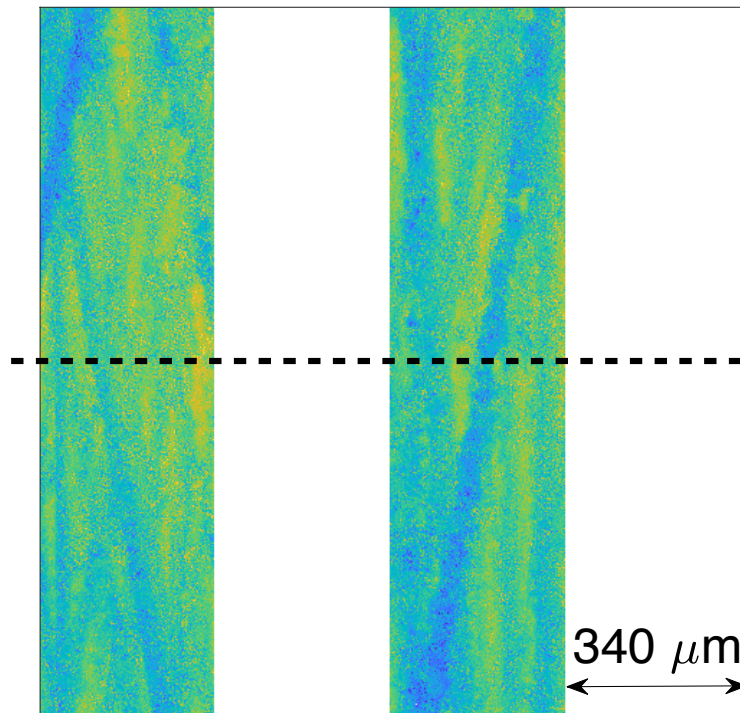
We use a Solarius confocal microscope to 3D scan large areas of the pad surface



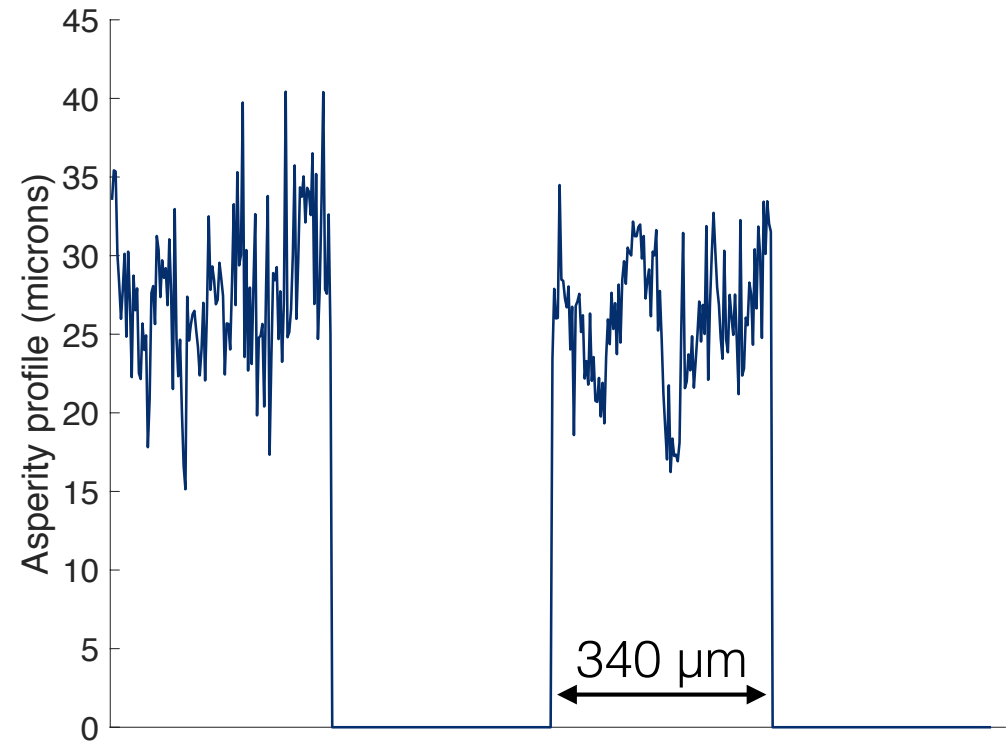
- Solarius allows for scanning large areas with a 25 μm resolution

[1] Brian Salazar, et al., "Die-scale modeling of planarization efficiency using segmented CMP pads: analyzing the effects of asperity topography" presented at the International Conference on Planarization/CMP Technology (ICPT), Hsinchu, Taiwan, September 2019.

We use a 3D laser scanning confocal microscope to scan small portions of the pad surface at high resolution ($\sim 2.5 \mu\text{m}$)



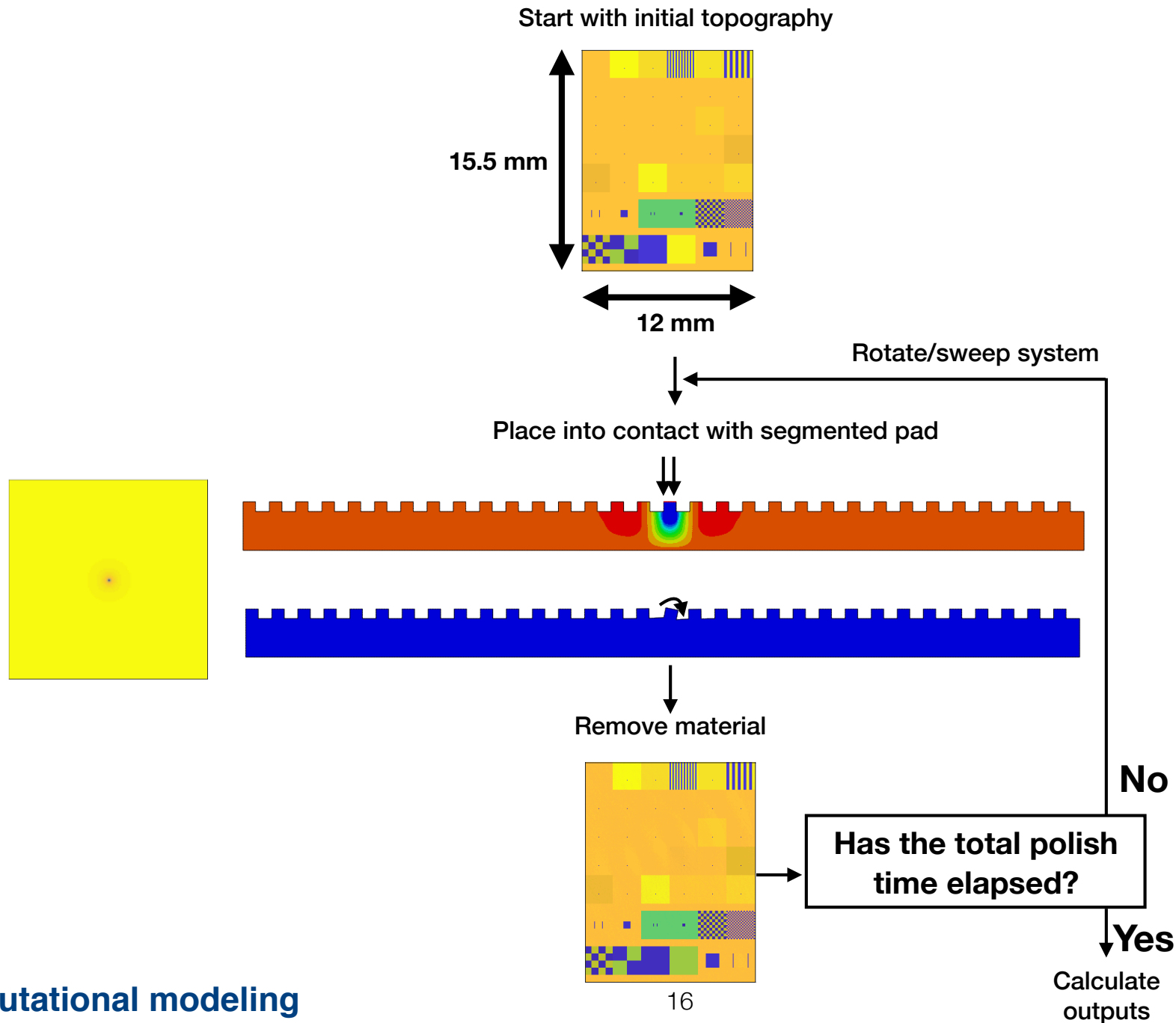
Pad Top View



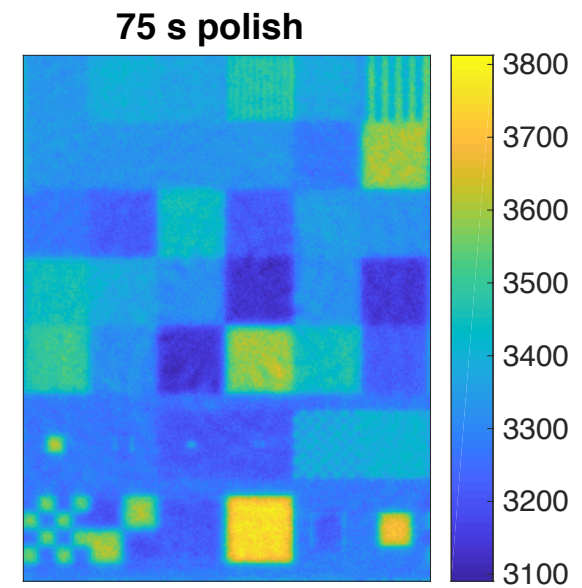
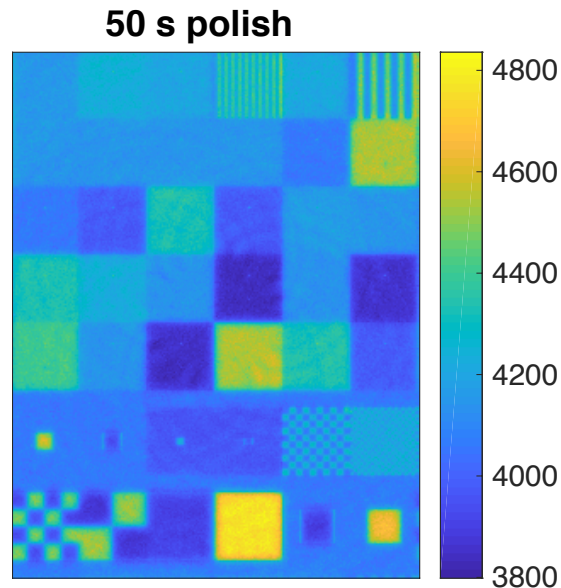
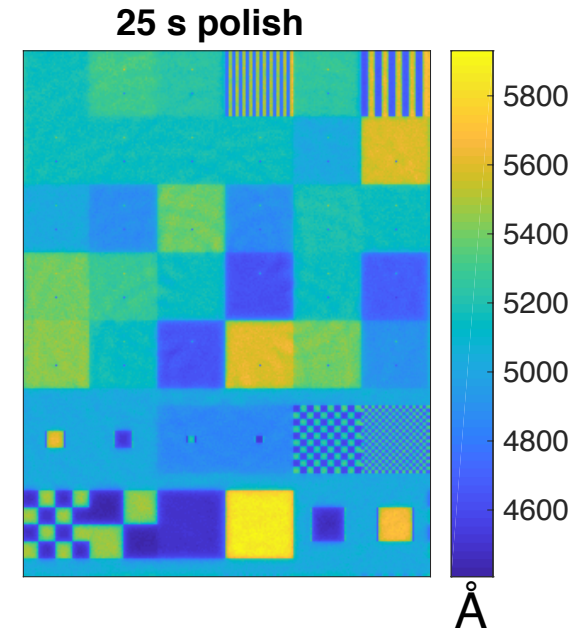
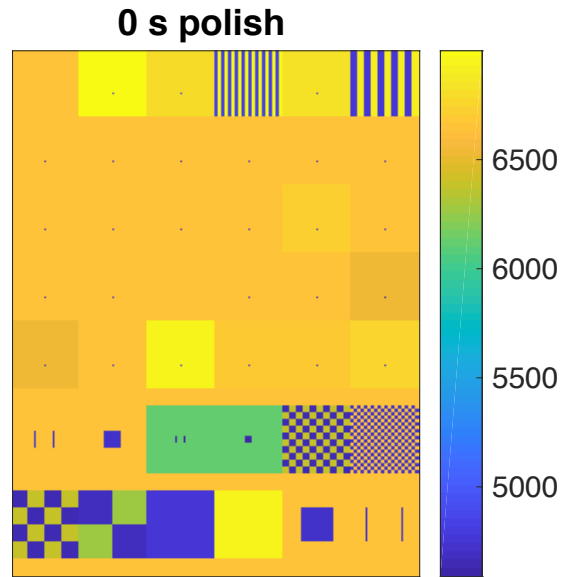
Pad Cross Section

[2] Brian Salazar and Hayden Taylor, "Computational modeling of segmented CMP pads; incorporating the effects of asperity topography" presented at the 22nd International Symposium on Chemical-Mechanical Planarization (CMP), Lake Placid, NY, August 2018.

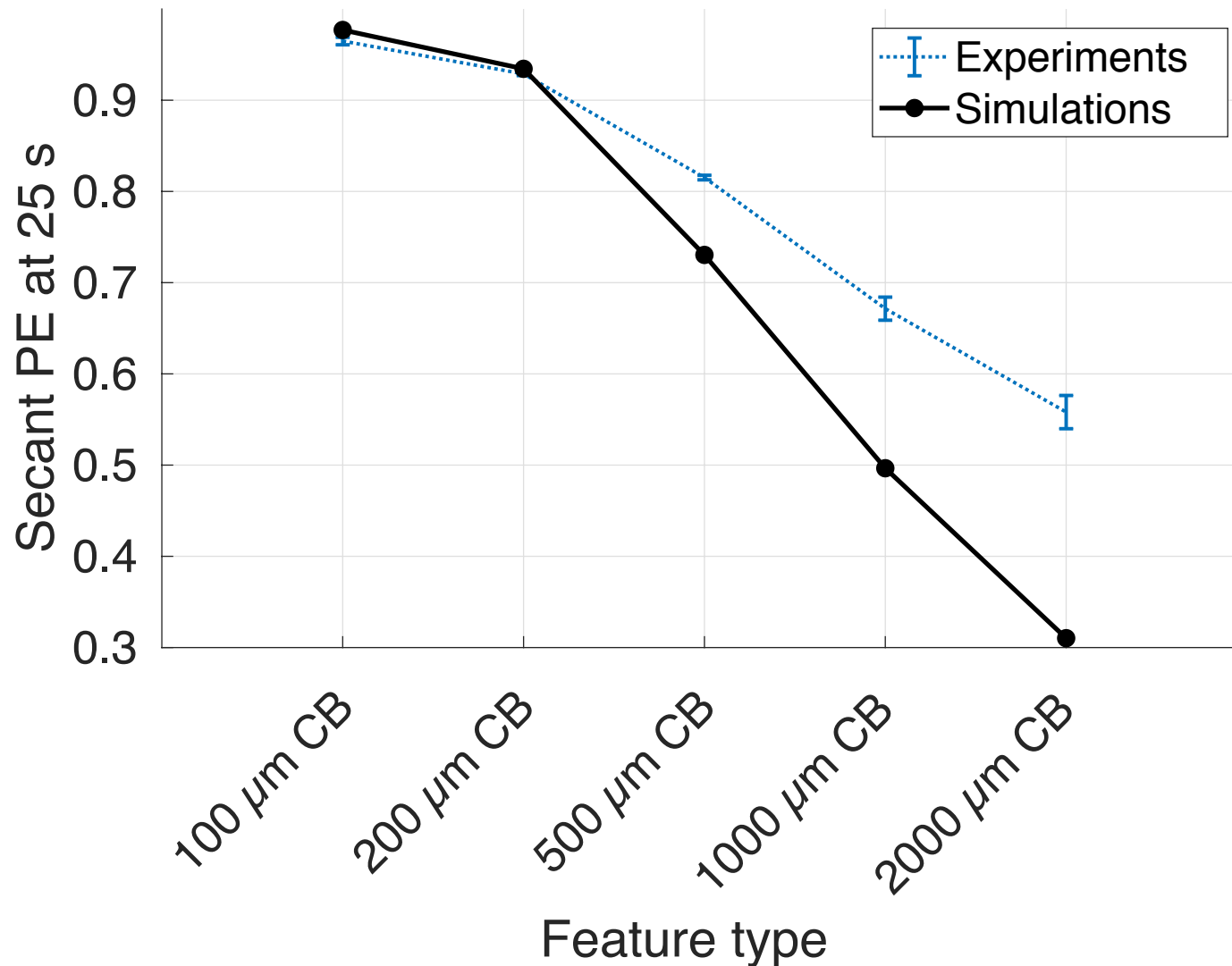
Material removal is predicted using the same framework, with a time-stepping approach and with the previously identified removal law



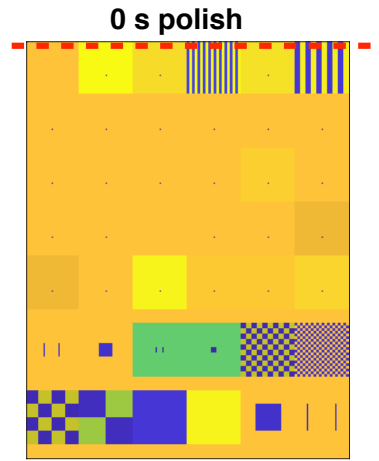
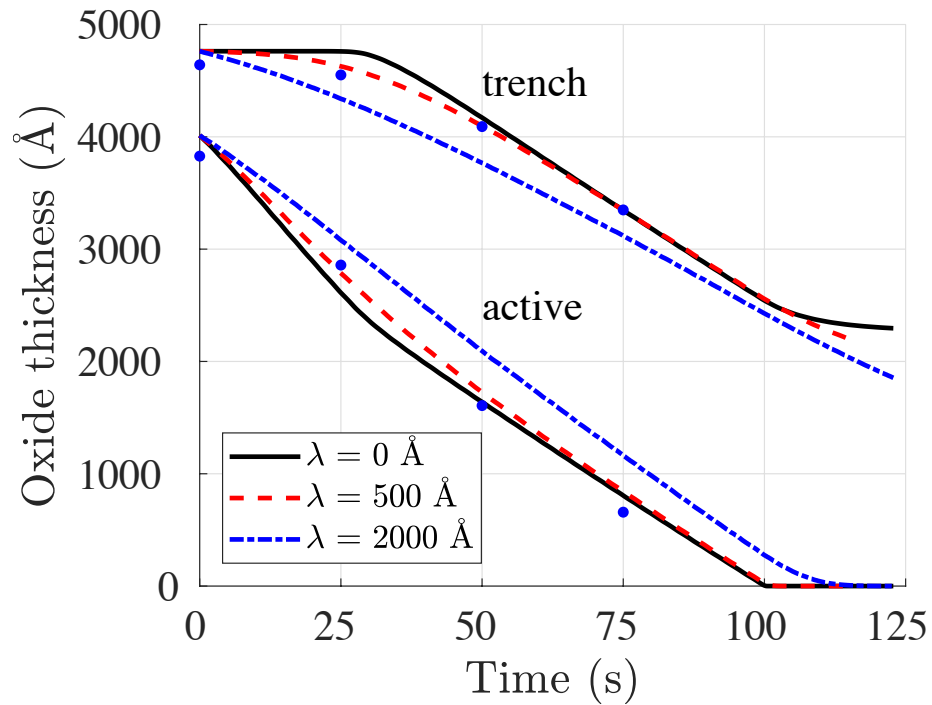
Simulated die topographies show the oxide becomes more planar as the polish continues



Simulations show consistent trends with experiments (i.e. feature size dependence). Predictions are more accurate for some features than others

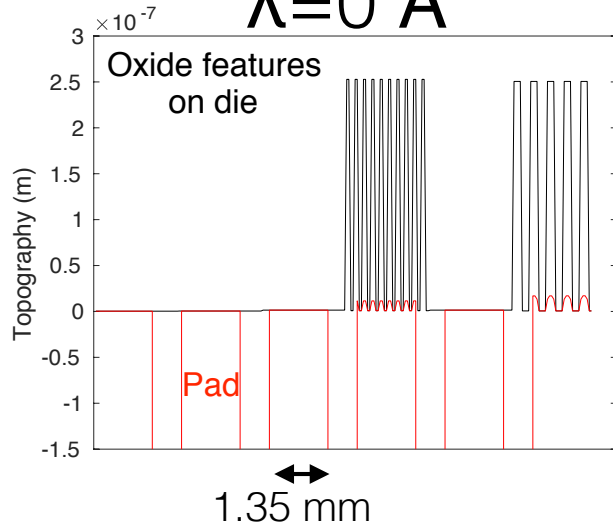


The model captures pad surface roughness effects

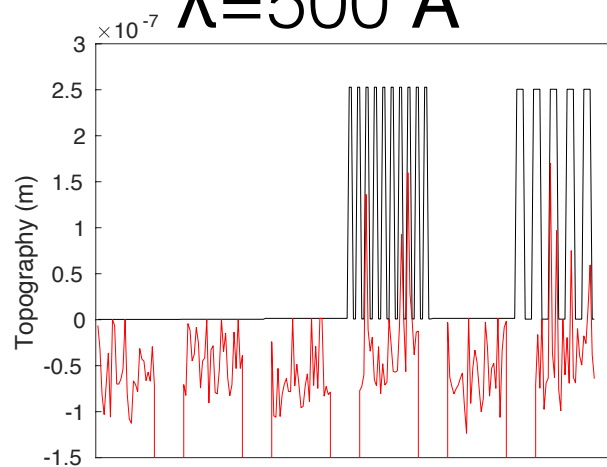


Top view

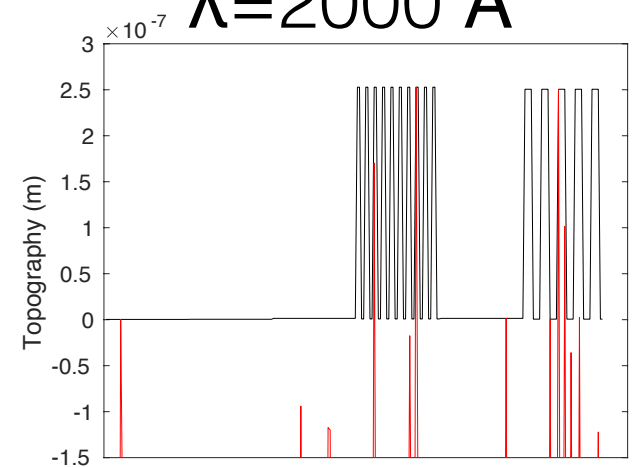
$\lambda = 0 \text{ Å}$



$\lambda = 500 \text{ Å}$

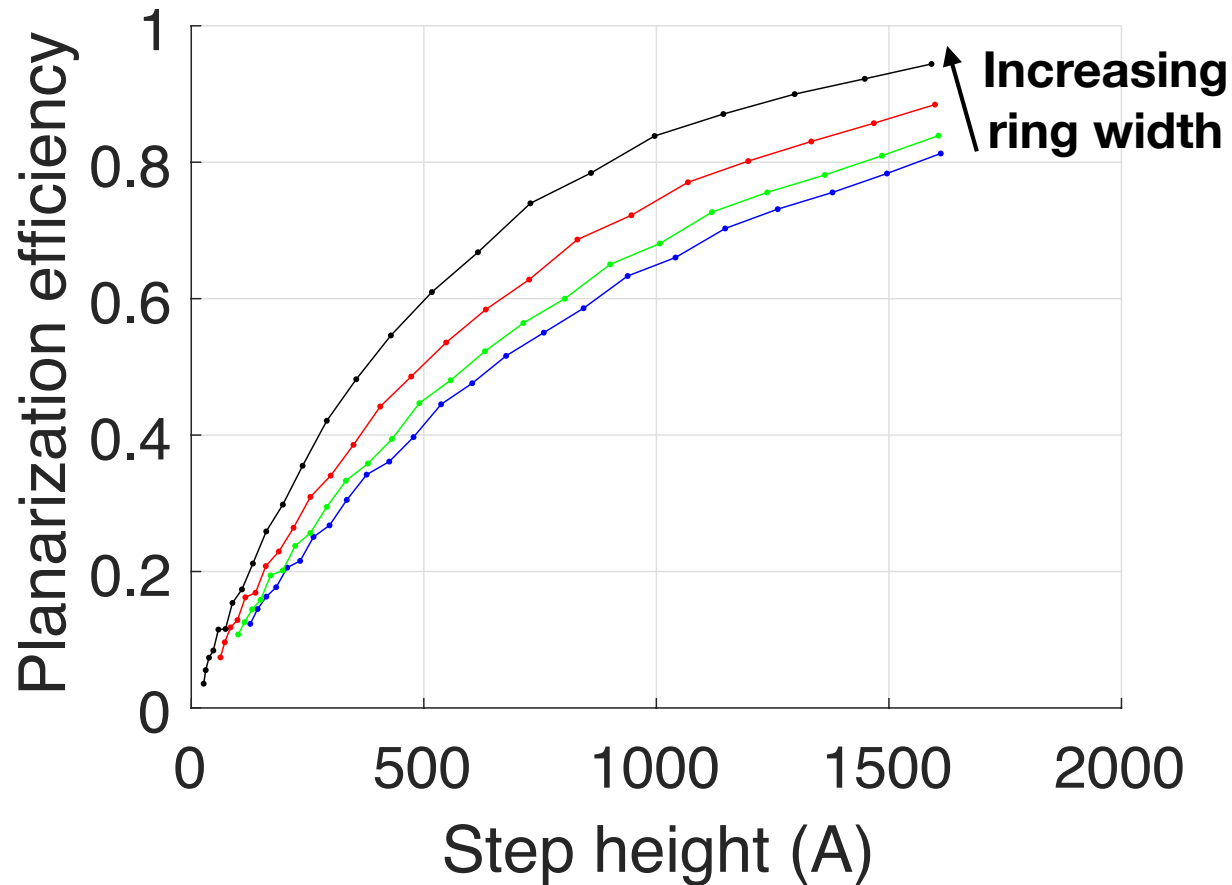
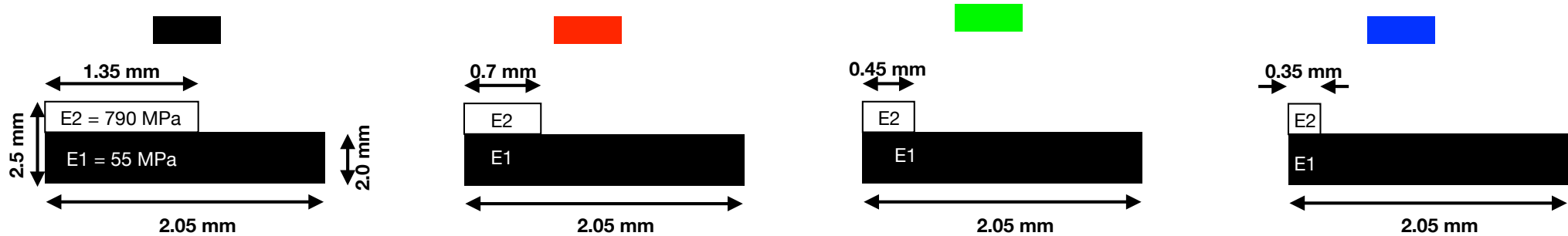


$\lambda = 2000 \text{ Å}$

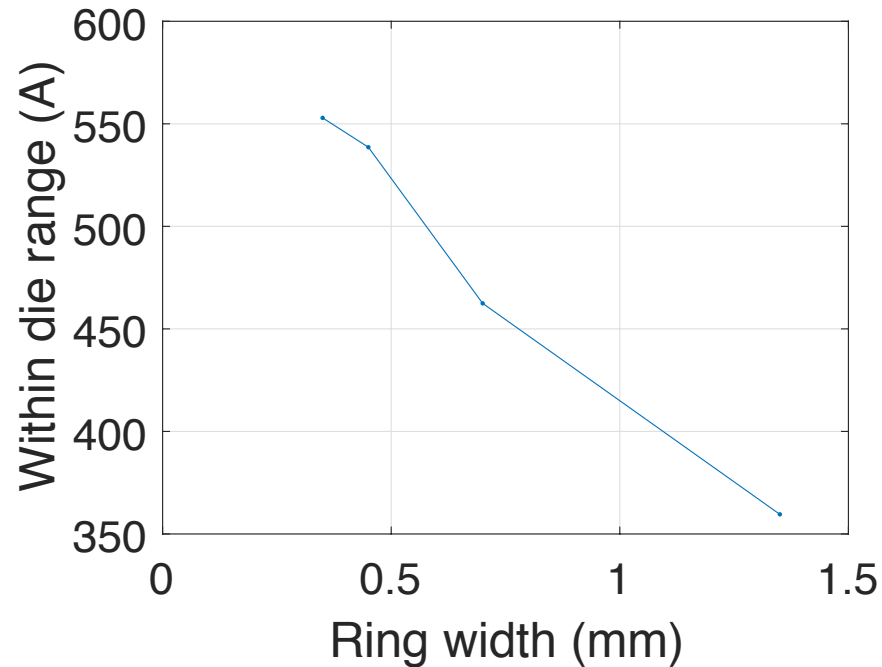


Cross-sectional views

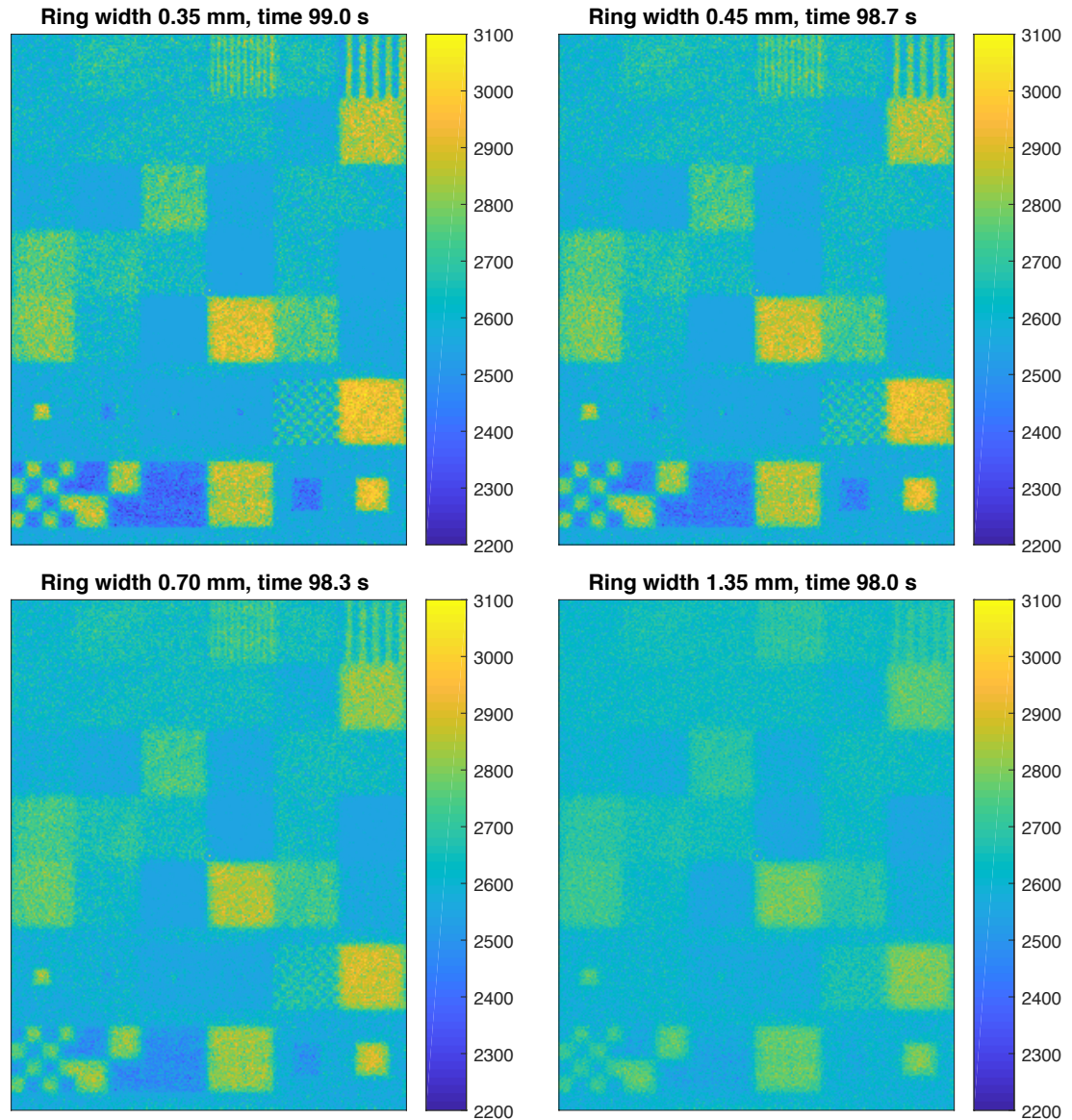
The model accurately predicts that rings with larger widths have higher planarization efficiencies



Pads with larger ring widths have smaller within-die active oxide thickness ranges



Topography (A)



Polish time is when the average active oxide thickness across the die is 100 Å

Results

Conclusion and Contributions

- We are able to identify material removal laws using simulated pressure distributions across a patterned die
- The pad surface texture is crucial for accurate prediction
- Using the identified material removal law, the die-scale model provides reasonable predictive ability.
- The die-scale model is able to capture the planarization efficiency's dependence on feature size